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STEP 4: BASELINE CONTROLLER PERFORMANCE SUMMARY

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Uncooled Max Temperature	: 59.27 °C
Baseline Max Temperature	: 36.58 °C
Total Active Cooling Energy	: 142.14 Wh
Total Cooling Runtime	: 602 seconds

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STEP 5: PREDICTIVE CONTROLLER PERFORMANCE SUMMARY

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Uncooled Max Temperature	: 59.27 °C
Step 4 Baseline Max Temp	: 36.58 °C
Step 5 Predictive Max Temp	: 33.60 °C
Baseline Active Energy	: 142.14 Wh
Predictive Active Energy	: 157.84 Wh
Energy Savings vs. Baseline	: -11.05 %

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Running Full Receding-Horizon MPC Simulation over 2001 steps...

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STEP 5: FULL MPC PERFORMANCE SUMMARY (COMPLETED)

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Uncooled Peak Temperature	: 59.27 °C
Step 4 Baseline Max Temp	: 36.58 °C
Step 5 Full MPC Max Temp	: 33.61 °C
Baseline Active Energy	: 142.14 Wh
Full MPC Active Energy	: 160.97 Wh
Energy Savings vs. Baseline	: -13.25 %

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Computing Step 6 Comparative Benchmarks & Advanced Degradation Metrics...

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STEP 6: COMPREHENSIVE THERMAL CONTROLLER BENCHMARK

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Metric	Uncooled	Step 4 Base	Step 5 Pred	Step 5 MPC
Peak Temperature (°C)	59.27	36.58	33.60	33.61
Time > 35°C Limit (s)	1491	313	0	0
Time > 32°C Setpoint (s)	1639	1634	1261	1426
Chiller Energy (Wh)	N/A	142.14	157.84	160.97
Energy Savings vs Base (%)	N/A	Baseline	-11.05	-13.25
Actuator Chatter (W/s)	N/A	6.00	23.00	4.41
Battery Aging Reduction (%)	N/A	Baseline	7.71	5.18
Added Range Gained (km)	N/A	0.000	-0.070	-0.084

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```
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UNIFIED EV THERMAL SIMULATION SUMMARY
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```
Total Duration           : 2000 s
Total Distance Travelled  : 10.216 km
Net Battery Energy Consumed : 2295.08 Wh
Peak Heat Generation      : 3772.83 W
Peak Ohmic Heat Loss       : 3752.46 W
Maximum Battery Temperature : 59.27 °C
Temperature Rise (Uncooled) : 34.27 °C
Initial SOC               : 90.00 %
Final SOC                 : 100.00 %
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```

```
[SUCCESS] All time-series objects exported to Base Workspace & EV_Thermal_Inputs.mat!
```

```
=====
Total Script Execution Time: 447.945 seconds
=====
```

```
>> projectRoot = ...
    'C:\Users\HP\Documents\Predictive Electric Vehicle Cooling';
```

```
load(fullfile(projectRoot, 'EV_Thermal_Inputs.mat'));
```

```
Warning: File "C:\Users\HP\Documents\Predictive Electric Vehicle
Cooling\source_powertrain\Mini Project 1 (Electric Vehicle Energy Consumption
Calculator)\EV_Predictive_Cooling_Plant.slx" is missing, which will prevent you from
saving this model properly. If the file moved, then close the model and open it from the
new
location.
```

```
> In Simulink.slx.extractFileForPart (line 11)
In ModelAdvisor.ExclusionEditor/load
In ModelAdvisor/ExclusionEditor
In ModelAdvisor.ExclusionEditor.getInstance
In ModelAdvisor.MdlAdvMenus.commonFcn>getMAExclusionMenus
In ModelAdvisor.MdlAdvMenus.commonFcn
In ModelAdvisor.MdlAdvMenus.contextMenu
In SLStudio.ContextMenus
```

```
Warning: File "C:\Users\HP\Documents\Predictive Electric Vehicle
Cooling\source_powertrain\Mini Project 1 (Electric Vehicle Energy Consumption
Calculator)\EV_Predictive_Cooling_Plant.slx" is missing, which will prevent you from
saving this model properly. If the file moved, then close the model and open it from the
new
location.
```

```
> In slcheck.getAdvisorFilterManager
In Advisor.ExclusionEditor/fetchDataFromBackend
In Advisor/ExclusionEditor
In Advisor/ExclusionEditorWindow/createSession
In Advisor/WindowBase/initiate
```

```
In Advisor/ExclusionEditorUIService/getExclusionEditor
In Advisor.getExclusionEditor
In ModelAdvisor.MdlAdvMenus.commonFcn>getMAExclusionMenus
In ModelAdvisor.MdlAdvMenus.commonFcn
In ModelAdvisor.MdlAdvMenus.contextMenu
In SLStudio.ContextMenus
Warning: File "C:\Users\HP\Documents\Predictive Electric Vehicle
Cooling\source_powertrain\Mini Project 1 (Electric Vehicle Energy Consumption
Calculator)\EV_Predictive_Cooling_Plant.slx" is missing, which will prevent you from
saving this model properly. If the file moved, then close the model and open it from the
new
location.
> In slcheck.getAdvisorFilterManager
In Advisor.ExclusionEditor/fetchDataFromBackend
In Advisor/ExclusionEditor
In Advisor/ExclusionEditorWindow/createSession
In Advisor/WindowBase/initiate
In Advisor/ExclusionEditorUIService/getExclusionEditor
In Advisor.getExclusionEditor
In ModelAdvisor.MdlAdvMenus.commonFcn>getMAExclusionMenus
In ModelAdvisor.MdlAdvMenus.commonFcn
In ModelAdvisor.MdlAdvMenus.contextMenu
In SLStudio.ContextMenus
Warning: File "C:\Users\HP\Documents\Predictive Electric Vehicle
Cooling\source_powertrain\Mini Project 1 (Electric Vehicle Energy Consumption
Calculator)\EV_Predictive_Cooling_Plant.slx" is missing, which will prevent you from
saving this model properly. If the file moved, then close the model and open it from the
new
location.
> In Simulink.slx.extractFileForPart (line 11)
In CloneDetector.Utils.getFilterFilePath
In CloneDetector/Exclusions/getCloneDetectionFilterManager
In CloneDetector/Exclusions/readExclusionsFromFilterService
In CloneDetector/Exclusions/retrieveExclusions
In CloneDetector/Exclusions
In ModelAdvisor.MdlAdvMenus.commonFcn>getMAExclusionMenus
In ModelAdvisor.MdlAdvMenus.commonFcn
In contextMenuCloneDetection
In SLStudio.ContextMenus
Warning: File "C:\Users\HP\Documents\Predictive Electric Vehicle
Cooling\source_powertrain\Mini Project 1 (Electric Vehicle Energy Consumption
Calculator)\EV_Predictive_Cooling_Plant.slx" is missing, which will prevent you from
saving this model properly. If the file moved, then close the model and open it from the
new
location.
> In CloneDetector/Exclusions/getCloneDetectionFilterManager
In CloneDetector/Exclusions/readExclusionsFromFilterService
In CloneDetector/Exclusions/retrieveExclusions
In CloneDetector/Exclusions
In ModelAdvisor.MdlAdvMenus.commonFcn>getMAExclusionMenus
In ModelAdvisor.MdlAdvMenus.commonFcn
```

```

In contextMenuCloneDetection
In SLStudio.ContextMenus
Warning: File "C:\Users\HP\Documents\Predictive Electric Vehicle
Cooling\source_powertrain\Mini Project 1 (Electric Vehicle Energy Consumption
Calculator)\EV_Predictive_Cooling_Plant.slx" is missing, which will prevent you from
saving this model properly. If the file moved, then close the model and open it from the
new
location.
>> load('Step3_Dynamic_Loads.mat')
>> load('EV_Thermal_Inputs.mat')
>> load('EV_Thermal_Inputs.mat')
Warning: File "C:\Users\HP\Documents\Predictive Electric Vehicle
Cooling\source_powertrain\Mini Project 1 (Electric Vehicle Energy Consumption
Calculator)\EV_Predictive_Cooling_Plant.slx" is missing, which will prevent you from
saving this model properly. If the file moved, then close the model and open it from the
new
location.
>> whos Q_heat_ts
      Name              Size              Bytes  Class              Attributes

      Q_heat_ts         1x1              32398  timeseries

>> Q_heat_zero = timeseries( ...
    zeros(size(Q_heat_ts.Data)), ...
    Q_heat_ts.Time);

Q_heat_zero.Name = 'Q_heat_zero';
>> T_K = T_batt_log.Data(:);
t      = T_batt_log.Time(:);

T_C = T_K - 273.15;

fprintf('Initial battery temperature: %.2f °C\n', T_C(1));
fprintf('Maximum battery temperature: %.2f °C\n', max(T_C));
fprintf('Final battery temperature: %.2f °C\n', T_C(end));
Unable to resolve the name 'T_batt_log.Data'.

>> whos T_batt_log
>>
>>
>> whos T_batt_log
>> whos T_batt_log
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>> whos
      Name              Size              Bytes  Class
Attributes

```

A	1x1	8	double
A_pre	1x1	8	double
Batt_Cap_Ah	1x1	8	double
Batt_capacity	1x1	8	double
COP_chiller	1x1	8	double
C_th	1x1	8	double
Cd	1x1	8	double
Crr	1x1	8	double
E_a	1x1	8	double
E_base_Wh	1x1	8	double
E_batt_net	1x1	8	double
E_cooling_MPC_Wh	1x1	8	double
E_cooling_baseline_Wh	1x1	8	double
E_cooling_pred_Wh	1x1	8	double
E_cum_base	2001x1	16008	double
E_cum_mpc	2001x1	16008	double
E_cum_pred	2001x1	16008	double
E_drive_Wh	1x1	8	double
E_mpc_Wh	1x1	8	double
E_pred_Wh	1x1	8	double
E_regen_Wh	1x1	8	double
E_savings_percent	1x1	8	double
E_step_Wh	1x1	8	double
E_traction_Wh	1x1	8	double
F_Rint	1x1	752	griddedInterpolant
F_acc	2001x1	16008	double
F_aero	2001x1	16008	double
F_rr	1x1	8	double
F_total	2001x1	16008	double
H	1x1	8	double
H_p	1x1	8	double
I_batt	2001x1	16008	double
I_batt_ts	1x1	32398	timeseries
I_charge_demand	2001x1	16008	double
I_charge_profile	900x1	7200	double
K	1x1	8	double
MPC_energy_savings	1x1	8	double
N	1x1	8	double
OCV_curve	1x11	88	double
P_accessory	1x1	8	double
P_avail	1x1	8	double
P_batt_char_max	1x1	8	double
P_batt_disch_max	1x1	8	double
P_batt_drive	2001x1	16008	double
P_cov	1x1	8	double
P_drive	2001x1	16008	double
P_mech	2001x1	16008	double
P_motor_max	1x1	8	double
P_pump	1x1	8	double
P_regen	2001x1	16008	double

P_regen_limit	1x1	8	double
P_term	1x1	8	double
P_wheel	2001x1	16008	double
Q_cooling_MPC	2001x1	16008	double
Q_cooling_baseline	2001x1	16008	double
Q_cooling_max	1x1	8	double
Q_cooling_predictive	2001x1	16008	double
Q_dist	120x1	960	double
Q_entropic	2001x1	16008	double
Q_heat	2001x1	16008	double
Q_heat_ts	1x1	32398	timeseries
Q_heat_zero	1x1	32401	timeseries
Q_ohmic	2001x1	16008	double
Q_passive	1x1	8	double
Q_passive_cooling	1x1	8	double
Q_pred_future	2001x1	16008	double
Q_proc	1x1	8	double
Q_req	1x1	8	double
Q_thresh	1x1	8	double
R_gas	1x1	8	double
R_i_map	5x5	200	double
R_internal_vec	2001x1	16008	double
R_meas	1x1	8	double
R_th	1x1	8	double
SOC	2001x1	16008	double
SOC_curr	1x1	8	double
SOC_curve	1x11	88	double
SOC_est	2001x1	16008	double
SOC_init	1x1	8	double
SOC_pred	1x1	8	double
SOC_ts	1x1	32392	timeseries
SOC_vec	1x5	40	double
T_a_dist	120x1	960	double
T_amb	2001x1	16008	double
T_amb_ts	1x1	32396	timeseries
T_avail	1x1	8	double
T_batt	2001x1	16008	double
T_batt_MPC	2001x1	16008	double
T_batt_baseline	2001x1	16008	double
T_batt_predictive	2001x1	16008	double
T_curr	1x1	8	double
T_kelvin	1x1	8	double
T_max	1x1	8	double
T_max_limit	1x1	8	double
T_motor_max	1x1	8	double
T_off	1x1	8	double
T_on	1x1	8	double
T_peak_base	1x1	8	double
T_peak_mpc	1x1	8	double
T_peak_pred	1x1	8	double
T_peak_uncooled	1x1	8	double

T_pre_target	1x1	8	double
T_regen_max	1x1	8	double
T_req	1x1	8	double
T_rise	1x1	8	double
T_set	1x1	8	double
T_set_predictive	1x1	8	double
T_vec	1x5	40	double
Tn	1x1	8	double
V_batt	2001x1	16008	double
V_meas	2001x1	16008	double
V_nom	1x1	8	double
V_oc_pred	1x1	8	double
V_oc_vec	2001x1	16008	double
a	2001x1	16008	double
a_max	1x1	8	double
a_min	1x1	8	double
aging_base	1x1	8	double
aging_mpc	1x1	8	double
aging_pred	1x1	8	double
aging_uncooled	1x1	8	double
cooling_active_state	2001x1	16008	double
cost_fun	1x1	32	function_handle
dT	1x1	8	double
dU_dT	1x1	8	double
d_to_charger	1x1	8	double
delta	1x1	8	double
dist_km	1x1	8	double
dist_total_km	1x1	8	double
drive_eff	2001x1	16008	double
dt	1x1	8	double
elapsed_time	1x1	8	double
est_cycles_base	1x1	8	double
est_cycles_mpc	1x1	8	double
eta_drivetrain	1x1	8	double
eta_tire	1x1	8	double
g	1x1	8	double
gear_ratio	1x1	8	double
horizon_len	1x1	8	double
i	1x1	8	double
idx_end	1x1	8	double
idx_throttle_end	1x1	8	double
incoming_charge_flag	1x1	1	logical
is_cooling_on	1x1	1	logical
k	1x1	8	double
k_deg_base	2001x1	16008	double
k_deg_mpc	2001x1	16008	double
k_deg_pred	2001x1	16008	double
k_deg_uncooled	2001x1	16008	double
lb	120x1	960	double
life_ext_mpc_pct	1x1	8	double
life_ext_pred_pct	1x1	8	double

m	1x1	8	double
m_eq	1x1	8	double
noise	2001x1	16008	double
nonlcon_fun	1x1	32	function_handle
omega	2001x1	16008	double
omega_base	1x1	8	double
omega_max	1x1	8	double
omega_wheel	2001x1	16008	double
on	1x1	8	double
opts	1x1	72237	optim.options.Fmincon
out	1x1	76478	Simulink.SimulationOutput
peak_throttle_lookahead	1x1	8	double
projectRoot	1x57	114	char
r_wheel	1x1	8	double
range_gain_mpc_km	1x1	8	double
range_gain_pred_km	1x1	8	double
regen_eff	2001x1	16008	double
rho	1x1	8	double
s_charger	1x1	8	double
s_route	2001x1	16008	double
savings_mpc_pct	1x1	8	double
savings_pred_pct	1x1	8	double
spec_cons_Wh_km	1x1	8	double
t	2001x1	16008	double
t_charge	900x1	7200	double
t_drive	601x1	4808	double
t_to_charger	1x1	8	double
tau_control	1x1	8	double
theta_throttle	2001x1	16008	double
time_above_32_base	1x1	8	double
time_above_32_mpc	1x1	8	double
time_above_32_pred	1x1	8	double
time_above_35_base	1x1	8	double
time_above_35_mpc	1x1	8	double
time_above_35_pred	1x1	8	double
time_above_35_uncooled	1x1	8	double
u_init	120x1	960	double
u_opt	120x1	960	double
u_prev	1x1	8	double
ub	120x1	960	double
v	2001x1	16008	double
v_curr	1x1	8	double
v_drive	601x1	4808	double
v_min_regen	1x1	8	double
v_ts	1x1	32388	timeseries
w_E	1x1	8	double
w_T	1x1	8	double
w_dU	1x1	8	double
wear_base	1x1	8	double
wear_mpc	1x1	8	double
wear_pred	1x1	8	double

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```
>> T_K = out.T_batt_log.Data(:);  
t      = out.T_batt_log.Time(:);
```

```

T_C = T_K - 273.15;

fprintf('Initial battery temperature: %.2f °C\n', T_C(1));
fprintf('Maximum battery temperature: %.2f °C\n', max(T_C));
fprintf('Final battery temperature: %.2f °C\n', T_C(end));
Initial battery temperature: 26.85 °C
Maximum battery temperature: 35.51 °C
Final battery temperature: 35.51 °C
>>
>>
>>
>> time_above_32 = trapz(t, double(T_C >= 32));
time_above_35 = trapz(t, double(T_C >= 35));

fprintf('Time above 32 °C: %.1f s\n', time_above_32);
fprintf('Time above 35 °C: %.1f s\n', time_above_35);
Time above 32 °C: 539.3 s
Time above 35 °C: 162.7 s
>>
>>
>>
>> fprintf('Maximum baseline cooling power: %.2f W\n', max(Q_cooling_baseline));
fprintf('Maximum MPC cooling power: %.2f W\n', max(Q_cooling_MPC));
fprintf('Maximum predictive cooling power: %.2f W\n', max(Q_cooling_predictive));
Maximum baseline cooling power: 2000.00 W
Maximum MPC cooling power: 2000.00 W
Maximum predictive cooling power: 2000.00 W
>>
>>
>>
>>
>>
>> fprintf('Baseline cooling active samples: %d\n', ...
    sum(cooling_active_state > 0));
Baseline cooling active samples: 602
>>
>>
>>
>>
>> fprintf('Maximum battery temperature: %.2f °C\n', max(T_C));
fprintf('Temperature at 32 °C threshold: %.2f °C\n', T_C(find(T_C >= 32,1)));
Maximum battery temperature: 35.51 °C
Temperature at 32 °C threshold: 32.01 °C
>>
>>
>> fprintf('Maximum battery heat generation: %.2f W\n', max(Q_heat));
fprintf('Average battery heat generation: %.2f W\n', mean(Q_heat));
fprintf('Maximum baseline cooling: %.2f W\n', max(Q_cooling_baseline));
fprintf('Average baseline cooling: %.2f W\n', mean(Q_cooling_baseline));

```

```

Maximum battery heat generation: 3772.83 W
Average battery heat generation: 777.41 W
Maximum baseline cooling: 2000.00 W
Average baseline cooling: 601.70 W
>>
>>
>>
>> fprintf('Maximum battery heat generation: %.2f W\n', max(Q_heat));
fprintf('Average battery heat generation: %.2f W\n', mean(Q_heat));
fprintf('Maximum baseline cooling: %.2f W\n', max(Q_cooling_baseline));
fprintf('Average baseline cooling: %.2f W\n', mean(Q_cooling_baseline));
Maximum battery heat generation: 3772.83 W
Average battery heat generation: 777.41 W
Maximum baseline cooling: 2000.00 W
Average baseline cooling: 601.70 W
>>
>>
>>
>>
>>
>> figure;
plot(t, Q_heat, 'LineWidth', 1.2);
hold on;
plot(t, Q_cooling_baseline, 'LineWidth', 1.2);
grid on;

xlabel('Time (s)');
ylabel('Heat / Cooling Power (W)');
legend('Battery Heat Generation','Baseline Cooling');
title('Battery Heat Generation vs Reactive Cooling');

```

Error using plot

Specify the coordinates as vectors or matrices of the same size, or as a vector and a matrix that share the same length in at least one dimension.

```

>>
>>
>> t_Q = out.tout(1:length(Q_heat));
>>
>>
>>
>> size(t_Q)
size(Q_heat)
size(Q_cooling_baseline)

ans =

```

```
ans =  
  
    2001         1  
  
ans =  
  
    2001         1  
  
>>  
>>  
>>  
>>  
>>  
>> figure;  
  
plot(t_Q, Q_heat, 'LineWidth', 1.2);  
hold on;  
plot(t_Q, Q_cooling_baseline, 'LineWidth', 1.2);  
  
grid on;  
  
xlabel('Time (s)');  
ylabel('Power (W)');  
legend('Battery Heat Generation','Baseline Cooling');  
title('Battery Heat Generation vs Reactive Cooling');  
  
>>  
>>  
>>  
>>  
>>  
>> Q_excess = Q_heat - Q_cooling_baseline;  
  
fprintf('Maximum excess heat: %.2f W\n', max(Q_excess));  
fprintf('Average excess heat: %.2f W\n', mean(Q_excess));  
  
time_heat_exceeds_cooling = trapz(t_Q, double(Q_heat > Q_cooling_baseline));  
  
fprintf('Time Q_heat > Q_cooling: %.1f s\n', ...  
    time_heat_exceeds_cooling);  
Maximum excess heat: 3772.83 W  
Average excess heat: 175.71 W  
Time Q_heat > Q_cooling: 419.5 s  
>>  
>>  
>>  
>>  
>>
```

```
>> E_heat_Wh = trapz(t_Q, Q_heat) / 3600;
E_cool_base_Wh = trapz(t_Q, Q_cooling_baseline) / 3600;

fprintf('Total generated heat: %.2f Wh\n', E_heat_Wh);
fprintf('Total baseline cooling: %.2f Wh\n', E_cool_base_Wh);
Total generated heat: 196.58 Wh
Total baseline cooling: 167.20 Wh
>>
>>
>>
>>
>> time_cooling_exceeds_heat = trapz(t_Q, ...
    double(Q_cooling_baseline > Q_heat));

fprintf('Time Q_cooling > Q_heat: %.1f s\n', ...
    time_cooling_exceeds_heat);
Time Q_cooling > Q_heat: 228.5 s
>>
>>
>>
>>
>>
>> time_heat_exceeds_cooling = trapz(t_Q, ...
    double(Q_heat > Q_cooling_baseline));

fprintf('Time Q_heat > Q_cooling: %.1f s\n', ...
    time_heat_exceeds_cooling);
Time Q_heat > Q_cooling: 419.5 s
>>
>>
>>
>>
>> Q_excess_cooling = max(Q_cooling_baseline - Q_heat, 0);

E_excess_cooling_Wh = trapz(t_Q, Q_excess_cooling) / 3600;

fprintf('Cooling energy above instantaneous heat load: %.2f Wh\n', ...
    E_excess_cooling_Wh);
Cooling energy above instantaneous heat load: 67.53 Wh
>>
>>
>>
>>
>> fprintf('Minimum ambient temperature: %.2f °C\n', ...
    min(T_amb)-273.15);

fprintf('Maximum ambient temperature: %.2f °C\n', ...
    max(T_amb)-273.15);

fprintf('Initial ambient temperature: %.2f °C\n', ...
```

```
T_amb(1)-273.15);

fprintf('Final ambient temperature: %.2f °C\n', ...
    T_amb(end)-273.15);
Minimum ambient temperature: -248.15 °C
Maximum ambient temperature: -237.61 °C
Initial ambient temperature: -248.15 °C
Final ambient temperature: -237.61 °C
>>
>>
>>
>>
>>
>>
>> figure;

plot(t_Q, T_amb, 'LineWidth', 1.2);

grid on;
xlabel('Time (s)');
ylabel('Ambient Temperature (K)');
title('Ambient Temperature');
>>
>>
>>
>>
>>
>>
>> T_amb_ts.Data(1)
T_amb_ts.Data(end)

ans =

    298.1500

ans =

    308.6946

>>
>>
>>
>>
>> fprintf('T_amb_ts initial = %.4f\n', T_amb_ts.Data(1));
fprintf('T_amb_ts final   = %.4f\n', T_amb_ts.Data(end));
T_amb_ts initial = 298.1500
T_amb_ts final   = 308.6946
>>
>>
>>
```

```
>>
>> T_amb(1)
T_amb(end)

min(T_amb)
max(T_amb)

ans =

    25

ans =

    35.5446

ans =

    25

ans =

    35.5446

>>
>>
>>
>> fprintf('T_amb first: %.4f\n', T_amb(1));
fprintf('T_amb last:  %.4f\n', T_amb(end));
fprintf('T_amb min:   %.4f\n', min(T_amb));
fprintf('T_amb max:   %.4f\n', max(T_amb));
T_amb first: 25.0000
T_amb last:  35.5446
T_amb min:   25.0000
T_amb max:   35.5446
>>
>>
>>
>>
>>
>> fprintf('T_amb_ts first: %.4f\n', T_amb_ts.Data(1));
fprintf('T_amb_ts last:  %.4f\n', T_amb_ts.Data(end));
T_amb_ts first: 298.1500
T_amb_ts last:  308.6946
>>
>>
>>
>>
>>
```



```

>>
>>
>> % Battery temperature in Celsius
T_batt_C = out.T_batt_log.Data(:) - 273.15;

% Ambient temperature in Celsius
T_amb_C = T_amb(:);

% Temperature difference between battery and ambient
DeltaT = T_batt_C - T_amb_C;

fprintf('Initial battery-ambient ΔT: %.2f °C\n', DeltaT(1));
fprintf('Maximum battery-ambient ΔT: %.2f °C\n', max(DeltaT));
fprintf('Final battery-ambient ΔT: %.2f °C\n', DeltaT(end));
Arrays have incompatible sizes for this operation.

Related documentation

>>
>>
>>
>> % Battery temperature
T_batt_C = out.T_batt_log.Data(:);
T_batt_C = T_batt_C - 273.15;

% Battery temperature time
t_batt = out.T_batt_log.Time(:);

% Ambient temperature
T_amb_C = T_amb(:);

% Ambient temperature time
t_amb = out.tout(1:length(T_amb_C));

% Interpolate ambient temperature onto battery time points
T_amb_interp = interp1(t_amb, T_amb_C, t_batt, 'linear', 'extrap');

% Battery-to-ambient temperature difference
DeltaT = T_batt_C - T_amb_interp;

fprintf('Initial battery-ambient ΔT: %.2f °C\n', DeltaT(1));
fprintf('Maximum battery-ambient ΔT: %.2f °C\n', max(DeltaT));
fprintf('Final battery-ambient ΔT: %.2f °C\n', DeltaT(end));
Initial battery-ambient ΔT: 1.85 °C
Maximum battery-ambient ΔT: 1.85 °C
Final battery-ambient ΔT: -1.95 °C
>>
>>
>>
>>
>>
>>

```

```
>> figure;

plot(t_batt, DeltaT, 'LineWidth', 1.4);

yline(0, '--');

grid on;

xlabel('Time (s)');
ylabel('\DeltaT = T_{battery} - T_{ambient} (°C)');

title('Battery Temperature Rise Above Ambient');
>>
>>
>>
>>
>>
>> figure;

plot(t_batt, T_batt_C, 'LineWidth', 1.4);
hold on;
plot(t_batt, T_amb_interp, 'LineWidth', 1.4);

grid on;

xlabel('Time (s)');
ylabel('Temperature (°C)');

legend('Battery Temperature', 'Ambient Temperature');

title('Battery Temperature vs Ambient Temperature');
>>
>>
>>
>>
>>
>>
>>
>> figure;

plot(t_batt, DeltaT, 'LineWidth', 1.4);

yline(0, '--');

grid on;

xlabel('Time (s)');
ylabel('\DeltaT = T_{battery} - T_{ambient} (°C)');

title('Battery Temperature Rise Above Ambient');
>> %% FINAL BATTERY THERMAL DIAGNOSTICS
```

```
% Battery temperature
T_batt_C = out.T_batt_log.Data(:) - 273.15;
t_batt    = out.T_batt_log.Time(:);

% Ambient temperature from actual timeseries
T_amb_C_ts = T_amb_ts.Data(:) - 273.15;
t_amb      = T_amb_ts.Time(:);

% Common time interval
t_start = max(t_batt(1), t_amb(1));
t_end   = min(t_batt(end), t_amb(end));

idx_batt = t_batt >= t_start & t_batt <= t_end;

t_common = t_batt(idx_batt);
T_batt_common = T_batt_C(idx_batt);

% Interpolate ambient onto battery time
T_amb_common = interp1(t_amb, T_amb_C_ts, ...
    t_common, 'linear');

% Battery temperature rise above ambient
DeltaT = T_batt_common - T_amb_common;

%% RESULTS

fprintf('\n===== BATTERY THERMAL DIAGNOSTICS =====\n');

fprintf('Simulation duration: %.1f s\n', t_end - t_start);

fprintf('\nBattery:\n');
fprintf('  Initial temperature: %.2f °C\n', T_batt_common(1));
fprintf('  Maximum temperature: %.2f °C\n', max(T_batt_common));
fprintf('  Final temperature:   %.2f °C\n', T_batt_common(end));

fprintf('\nAmbient:\n');
fprintf('  Initial temperature: %.2f °C\n', T_amb_common(1));
fprintf('  Maximum temperature: %.2f °C\n', max(T_amb_common));
fprintf('  Final temperature:   %.2f °C\n', T_amb_common(end));

fprintf('\nBattery - Ambient:\n');
fprintf('  Initial ΔT: %.2f °C\n', DeltaT(1));
fprintf('  Maximum ΔT: %.2f °C\n', max(DeltaT));
fprintf('  Minimum ΔT: %.2f °C\n', min(DeltaT));
fprintf('  Final ΔT:   %.2f °C\n', DeltaT(end));

fprintf('\nHeat generation:\n');
fprintf('  Maximum: %.2f W\n', max(Q_heat));
fprintf('  Average: %.2f W\n', mean(Q_heat));
```

```
fprintf('\nBaseline cooling:\n');
fprintf('  Maximum: %.2f W\n', max(Q_cooling_baseline));
fprintf('  Average: %.2f W\n', mean(Q_cooling_baseline));

fprintf('\nHeat vs cooling:\n');
fprintf('  Time heat > cooling: %.1f s\n', ...
    trapz(t_Q, double(Q_heat > Q_cooling_baseline)));

fprintf('  Time cooling > heat: %.1f s\n', ...
    trapz(t_Q, double(Q_cooling_baseline > Q_heat)));

fprintf('=====\n');

===== BATTERY THERMAL DIAGNOSTICS =====
Simulation duration: 1369.0 s

Battery:
  Initial temperature: 26.85 °C
  Maximum temperature: 35.51 °C
  Final temperature:   35.51 °C

Ambient:
  Initial temperature: 25.00 °C
  Maximum temperature: 33.85 °C
  Final temperature:   33.85 °C

Battery - Ambient:
  Initial  $\Delta T$ : 1.85 °C
  Maximum  $\Delta T$ : 1.94 °C
  Minimum  $\Delta T$ : -1.05 °C
  Final  $\Delta T$ :   1.66 °C

Heat generation:
  Maximum: 3772.83 W
  Average: 777.41 W

Baseline cooling:
  Maximum: 2000.00 W
  Average: 601.70 W

Heat vs cooling:
  Time heat > cooling: 419.5 s
  Time cooling > heat: 228.5 s
=====
>>
```